CC5053NI Cloud Computing And IOT

50% Group Coursework

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GitHub Link: https://github.com/Sabinpabt23/Smart-Plant.git

I confirm that I understand my coursework needs to be submitted online via My Second Teacher Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded

Table of Contents

1. Introduction 1
1.1 Current Scenario 1
1.2 Problem Statement and Project as a Solution 2
1.3 Aim and Objective of the Project 2
2. Background 3
2.1 Description of Smart Plant Monitoring System 3
2.2 Design Diagrams 4
2.3 Flow Chart of System diagram 5
2.4 Block Diagram 6
2.5 Requirement Analysis 7
3. Development 15
4. Results and Findings 16
5. Future Works 17
6. Conclusion 18
7. Bibliography 19

Table of Figures

Figure 1 Hardware, Circuit and Schematic Diagram 4
Figure 2 Flowchart..... 5
Figure 3 Block Diagram 6
Figure 4 ESP32 8
Figure 5 DHT11 SENSOR 8
Figure 6 PIR MOTION SENSOR..... 9
Figure 7 SOIL MOISTURE SENSOR 10
Figure 8 RELAY 5V MODULE 11
Figure 9 TACTILE BUTTON 12
Figure 10 WATER PUMP 13
Figure 11 BREADBOARD 14

1. Introduction

The Smart Plant Monitoring System enables users to observe and gauge environmental variables, which can affect plant growth and health by monitoring and collecting this information to get an understanding of the environmental variables influencing plant health and adaptability as well as providing proper care for plants based on this information.

1.1 Current Scenario

Most agricultural practices in Nepal are based on traditional ways of doing things; most people in Nepal grow and care for their plants according to traditional methods used in smallholder agriculture. This includes the fact that a large percentage of households in Nepal have rooftop gardens; many of these households have a way of nurturing, raising, and growing plants through their rooftop gardens; and a great deal of smaller landholders in Nepal have farmland that relies on the observation of crops/vegetable growth and/or soil condition in order to determine the health and condition of their plants. The inability of the average farmer or plant caretaker to gain access to newer and modern technologies for monitoring the growth of their plants means they must rely on personal experience and visual evaluations in order to gain an understanding of their fields. Many factors can contribute to the difficulty of sustaining a healthy crop; among them, seasonal variations, irregular rainfall, and limited access to water resources.

1.2 Problem Statement and Project as a Solution

This project enables to continuously monitor your crop conditions using this system, which means you'll know what effect the soil and environment have on your crops in real time. The information you receive from this system will help inform when you need to take action regarding your crops. Additionally, the system will conserve water and improve the health of your crops, which will assist in addressing some of the challenges in Nepal related to management of water and unpredictable weather patterns. The system can be implemented by individuals on their terrace roofs, or by small scale farms and greenhouses.

1.3 Aim and Objective of the Project

The project main goal is the development of a plant care support system that continuously monitors soil and environmental factors. The project assist in improving plant maintenance practices by providing accurate and timely information to the user so they can take action based on their plants evolving care needs, mainly in circumstances where manual means of plant care are typically employed.

Objective

- To continuously investigate soil quality and surrounding environmental conditions that affect plant growth
- To reduce the extent of reliance on guesswork and manual observation when maintaining plants
- To enhance the effective use of water through active and accurate identification of plant water requirements
- To promote more healthy and consistent growth of plants
- To develop an easy-to-use and practical solution that can be used by households and or small-scale farmers in Nepal; and

2. Background

The background section of the Smart Plant Monitoring System describes the fundamental concepts upon which the Smart Plant Monitoring System's technology is based. The background also includes an explanation of how the Smart Plant Monitoring System was built and how its component parts interact with one another and what is necessary for the Smart Plant Monitoring System to be successfully implemented. The background section will also provide an understanding of how to interpret and utilize the Smart Plant Monitoring System's design and operational specifications.

Sensors are connected to a Microcontroller that serves as a central controller using output devices. Sensors collected The collected sensor data will include moisture level within Soil, temperature, humidity and motion, which will be sent to the ESP32 Microcontroller to process the data and present real-time displays on an LCD. Through WiFi, the microcontroller will transmit the data collected to the cloud, where it can be accessed from anywhere in the world. If at any time the moisture level falls below a predetermined threshold, the microcontroller will activate a Water Pump via the relay module to automatically water the plant. The overall system has the ability to continuously monitor and use water efficiently while enhancing overall plant care.

2.1 Description of Smart Plant Monitoring System

The Smart Plant Monitoring System allows for continuous monitoring of essential environmental conditions that affect the rate of growth of plants. The Smart Plant Monitoring System gathers data continuously from both the soil that surrounds the plant and the environment. All of this data is analyzed to produce the output that provides the user with a clear picture of the current condition of the plant at any time.

Sensors are the primary component of the Smart Plant Monitoring System. Data collected via sensors relating to the soil conditions and environmental conditions, are sent to a central control unit, where the data is interpreted, and meaningful output produced. The output can be seen visually, such as in the form of charts, or can serve as a basis for

additional actions, such as alerts to users that they may need to check electronic devices connected to the Smart Plant Monitoring System, or automated actions such as watering the plant or providing nutrients to the soil. This process of collecting, processing and providing users with an output based on the results of the data collection is referred to as the structured approach and ensures that the plant will always be monitored accurately and continuously, through the use of the Smart Plant Monitoring System.

2.2 Design Diagrams

Circuit, Hardware Diagram and Schematic

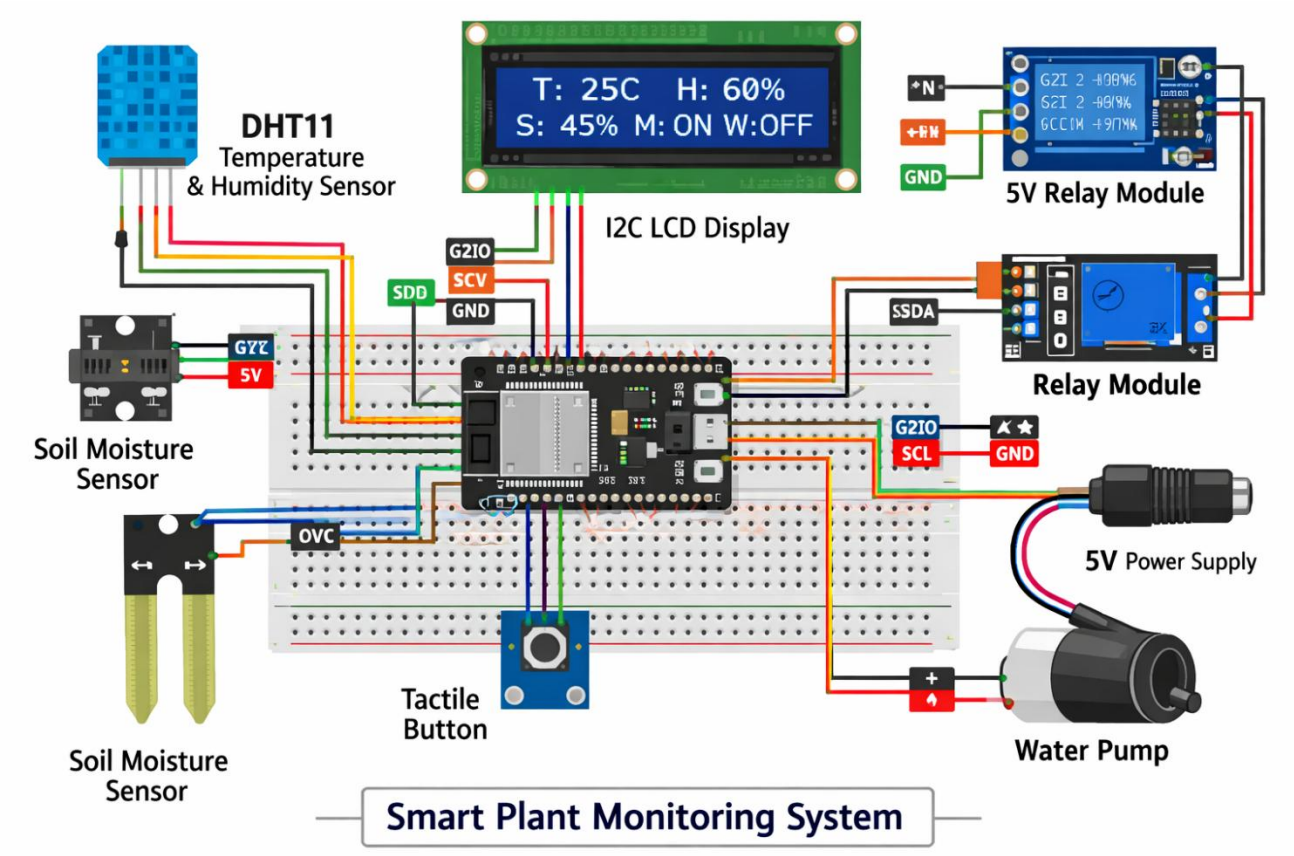

Figure 1 Hardware, Circuit and Schematic Diagram

2.3 Flow Chart of System diagram

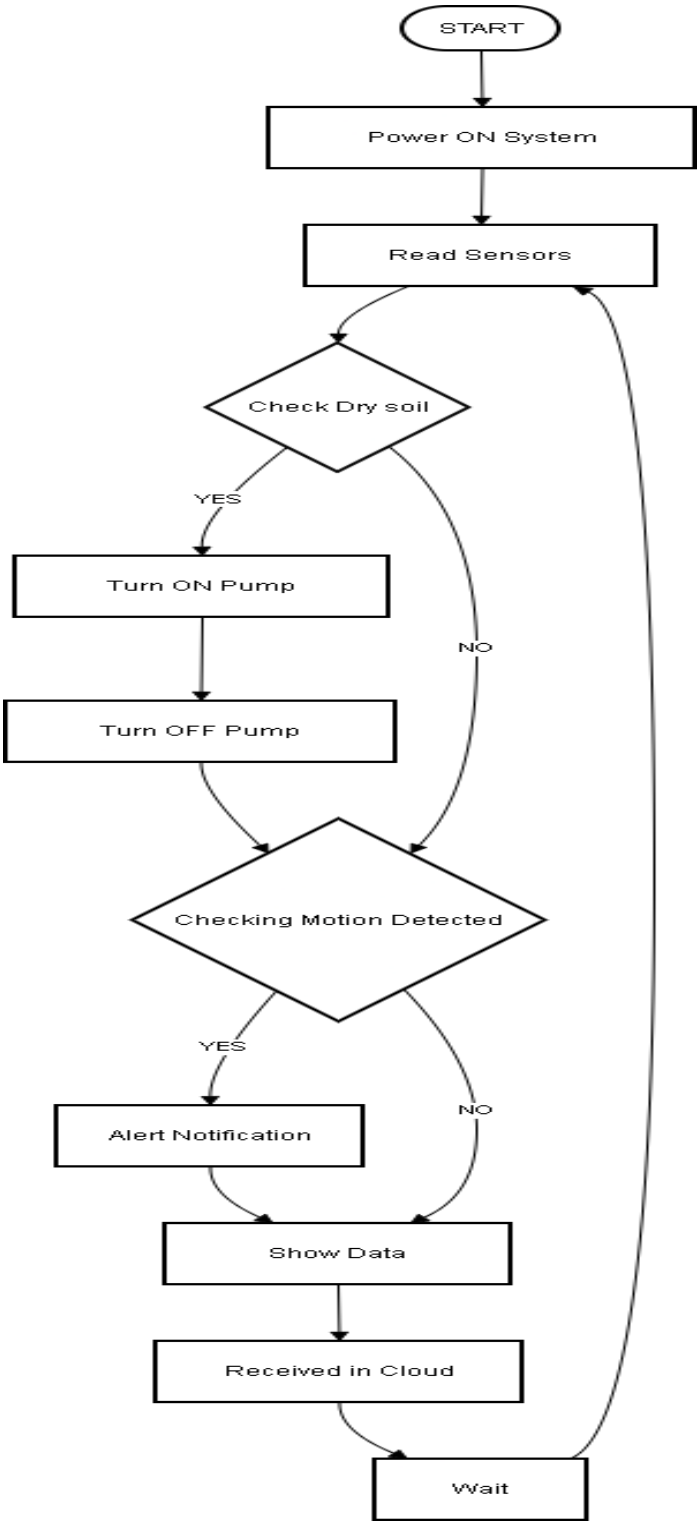

Figure 2 Flowchart

2.4 Block Diagram

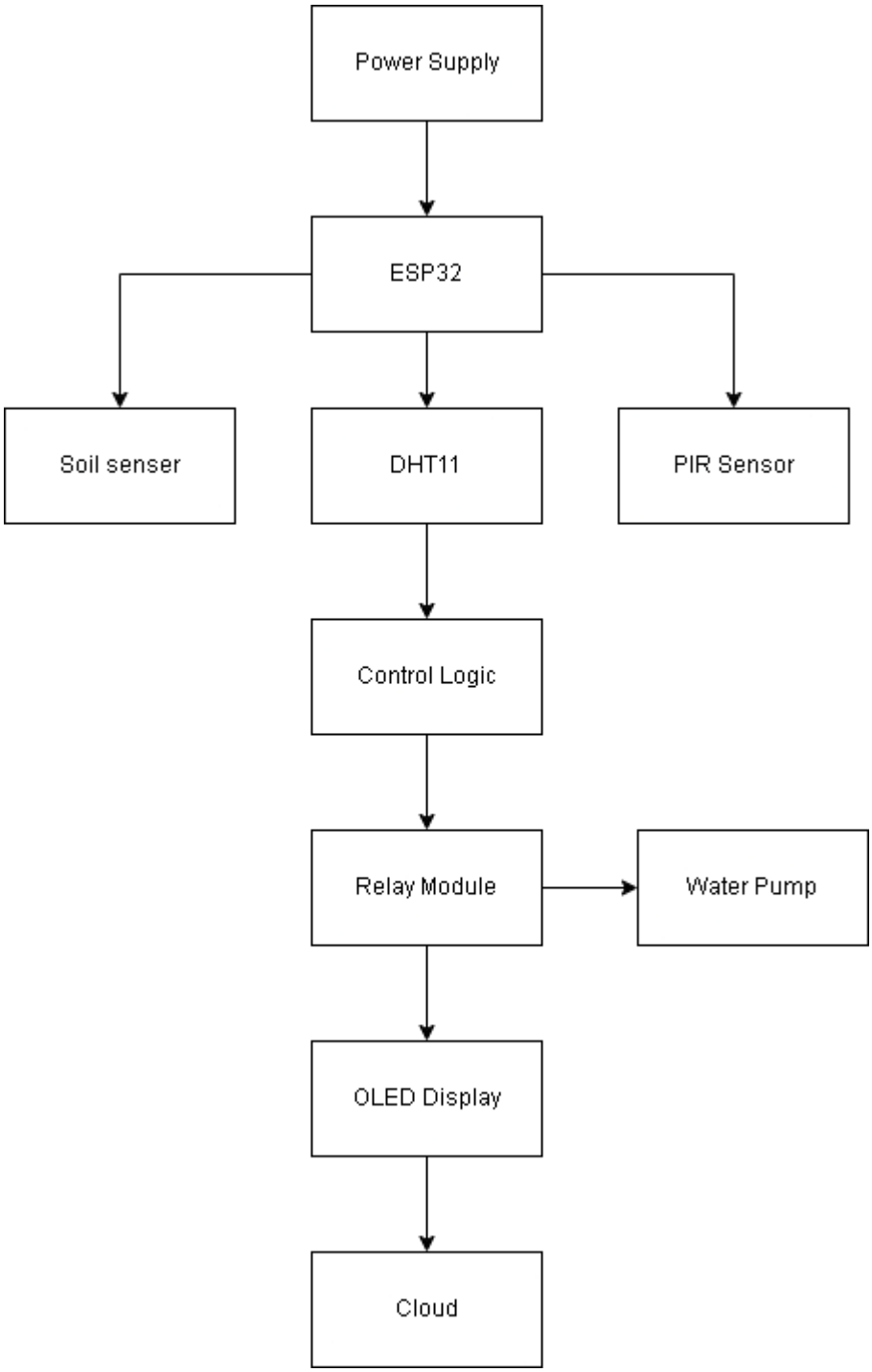

Figure 3 Block Diagram

2.5 Requirement Analysis

In the project of Smart Plant Monitoring System it is implemented and functions and provide impact in the environment through the Hardware parts. The parts and components used for this project are:

a) ESP32 (x1)

The Smart Plant Monitoring System Central Processing Unit is an ESP32 microcontroller. This microcontroller serves as the main controller of the entire project, allowing connections and control of all sensors, output devices, and the Internet of Things (IoT) platform.

For this particular project, the ESP32 will gather sensor information, process that information, create decisions based on that information, and interact with the Blynk Mobile App via Wi-Fi.

Why ESP32 is Chosen Over Other Microcontrollers

- The ESP32 contains an integrated Wi-Fi module, a feature needed in most Internet of Things (IoT) based applications.
- The ESP32 can simultaneously support multiple sensor types and also drive multiple output devices.
- The ESP32 supports both analogue and digital inputs and outputs, which makes it ideal for soil moisture, passive infrared (PIR), and digital humidity/temperature (DHT) sensors.
- The ESP32 is much faster than basic controllers such as an Arduino UNO in terms of processing speed.

Roles in the Project: In the project it is used as it is major choice as it allows to trigger data sensor, run the data flow and get all the necessary programming and commands running. This part helps in transferring data in the cloud with feature to monitor the process in mobile phones and in the web servers. It helps in controlling the water pump.

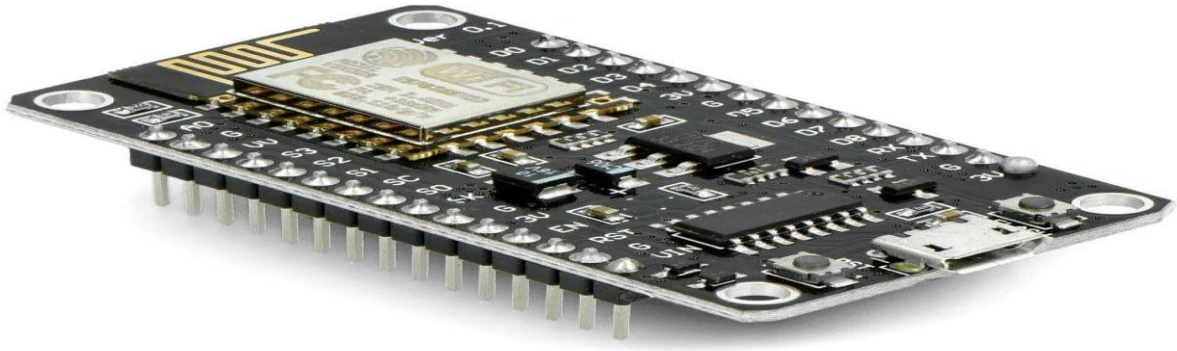

Figure 4 ESP32

b) DHT11 Sensor (x1)

Definitions: It is a sensor which measure temperature along with humidity which is digital has can temperature from 0 to 50 Celsius and calculate humidity from 20 to 80 percent. It is connected to GPIO 4 of the ESP32.

Roles in the Project: In the project it is used as in the environment all the plant need appropriate temperature and humidity for survival where DHT11 sensor is introduce to monitor current conditions of weather and summarize that whether Plant is suffering from High temperature and waterlessness which trigger the sensor and alert to add water.

Figure 5 DHT11 SENSOR

c) PIR Motion Sensor (x1)

Definitions: It is a sensor which measure radiation of heat by detecting motion of object from 5 to 7 meter. The sensor have presence of sensor name pyroelectric which help in transforming radiation to voltage resulting moving of devices like light, alarm. The sensor is connected in ESP where sensitivity can be adjusted.

Roles in the Project: In the project it is used in detecting certain change in movement like when Cows comes to consume Plant, the sensor detect and alerts proving security.

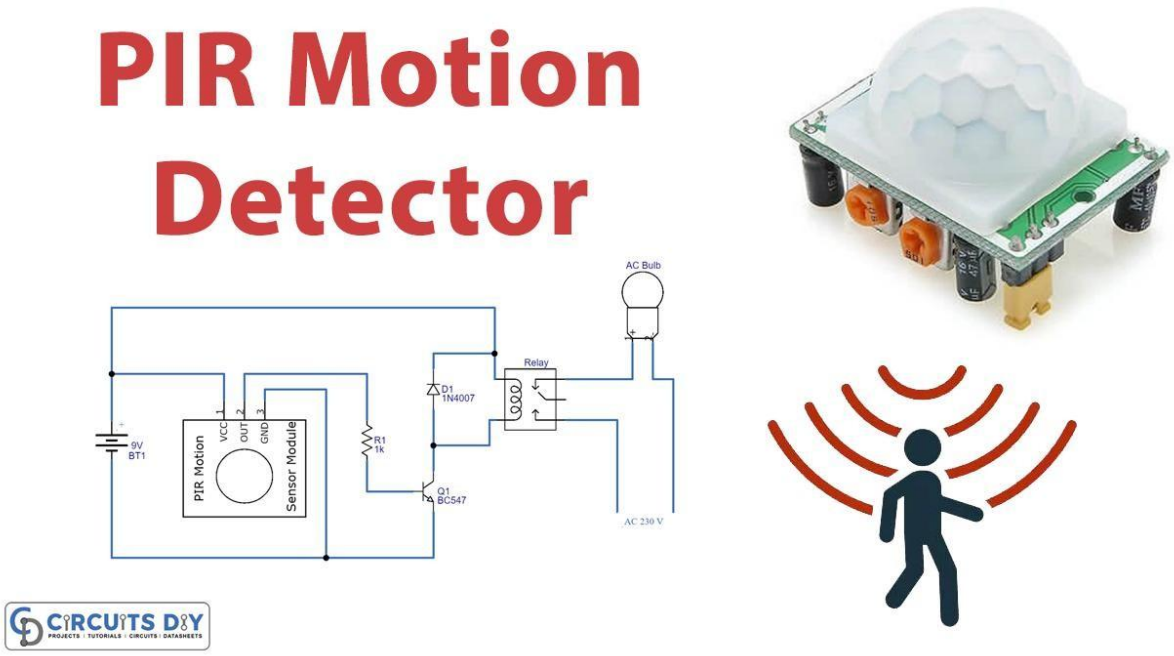

Figure 6 PIR MOTION SENSOR

d) Soil Moisture Sensor (x1)

Definitions: It is a sensor which measure Soil moisture that help to keep record of growth in plant and proper utilization of water in plant preventing over and under flow of water. The sensor is connected to GPIO 32 (analog pin) of the ESP32.

Roles in the Project: In the project it is used in judging health of plant by when soil is dry provides appropriate water to the plant with the help of water pump as this sensor is very popular in sector of agriculture.

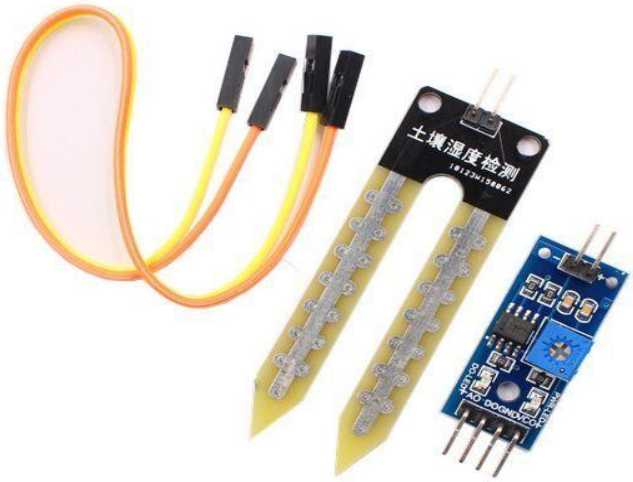

Figure 7 SOIL MOISTURE SENSOR

e) Relay 5V Module (x1)

Definitions: It is a switch of electromechanical which help in low power ESP32 microcontroller to function a higher power components such as motor, light and for the project water pump.

Roles in the Project: In the project it is used for the pump as it needs more power and ESP can directly match the voltage where Relay 5V Module is introduced as facilitator which helps pump to turn on and off with detection from Soil Moisture Sensor.

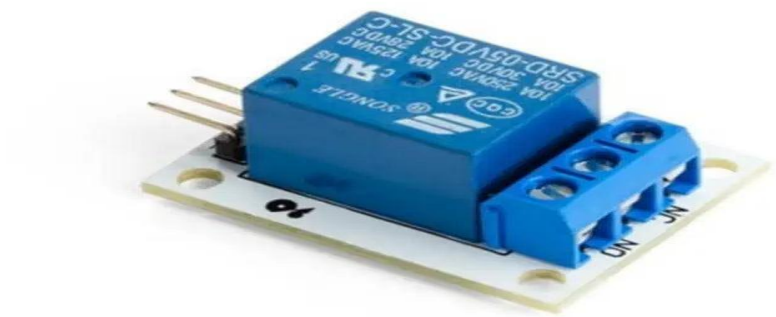

Figure 8 RELAY 5V MODULE

f) Tactile Button (x1)

Definitions: It is a button when clicked it indicates activation of circuit which allows a manual functioning of devices like remote and phone.

Roles in the Project: In the project it is used to press tactile button which allows water pump to run and throw water to the plants.

Figure 9 TACTILE BUTTON

g) Water Pump (x1)

Definitions: It is a pump which help in flow of water from one place to another. The pump is used in providing water to plants and soil. The Pump is provided appropriate power from Relay 5V Module.

Roles in the Project: In the project it is used to provide water to the plants when the sensors feels the soil lacks water which is important for conserving water and avoiding overflow of water

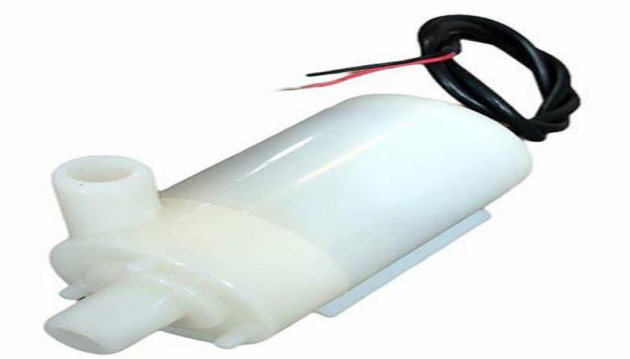

Figure 10 WATER PUMP

h) Breadboard (x2)

Definitions: It is a board which is one of foundation in creating of sample or prototype of devices which holds wire temporarily in holes which are inter-associated in the board. It is used in connecting batteries and wires.

Roles in the Project: In the project it is used to test circuits which is useful to make changes during the implementation and code developments

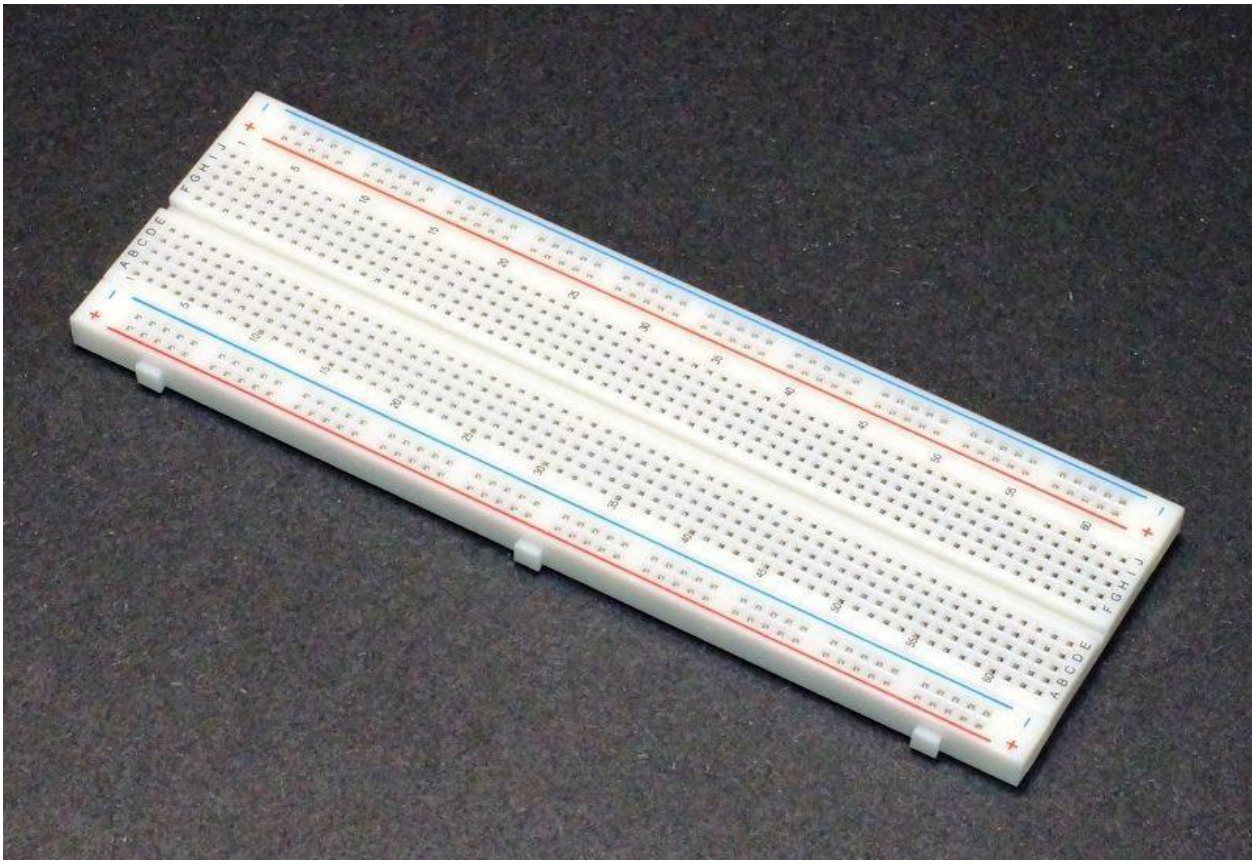

Figure 11 BREADBOARD

Additional Components used in Project are

- (3 - 7.4VOLT)18650 BATTERY x 1(3v) and x 1(7.4 v)
- JUMPER WIRE x 10(male to male) x 10(female to female) x10(male to female)

3. Development

The Smart Plant Monitoring System has been constructed in a manner that is systematic and coherent with the overall construction process, with the aim of producing reliable and real-time accurate readings from the sensors utilized in the system.

To achieve this goal, the following steps were taken during the system design:

a) Planning the project;

b) Selecting sensors and control units based upon their availability and costs, as well as how well they meet the needs of the environment in Nepal

c) Running preliminary tests to find out whether the sensors and control units functioned correctly prior to installing all components into the system.

After completing the tests for each component, they were assembled per the specifications outlined in circuit and architectural diagrams. In addition to assembling the components into one complete unit, I paid close attention to ensuring the power connection and proper sensor placement on the board were stable and accurate; if not, the data readings would be inaccurate. After completing the hardware assembly, I developed the control logic and uploaded this control logic to the host system.

The software component of the Smart Plant Monitoring System has been developed to continuously collect data from the sensors, process the collected data in real time, and create a meaningful representation of the data based on defined criteria that I set up during the software development process. After developing the software component, I ran several test scenarios using different environmental conditions to evaluate how it would respond to changes in the moisture content of the soil and also the temperature and humidity in the surrounding area. Issues detected during these test runs were corrected, and the tests were repeated and refined until every issue had been resolved.

4. Results and Findings

The reliable and consistent performance of the system was achieved through proper implementation of the Smart Plant Monitoring System. In particular, the accurate tracking of changes in soil condition as well as environmental influences over time, both of which contributed to producing these results. This was achieved through continuous monitoring, which allowed the user to better understand the different periods of water need (demand) as well as stress on the plant due to environmental influences (e.g. heat, cold).

The results from continuous monitoring demonstrate that providing accurate soil and environmental data leads to a greater understanding of how plants grow than did manual observations of soil and environmental conditions. This system was able to demonstrate to the user the true condition of their soil, which will lead to less water usage by informing them when their plants needed more water and when they did not. As water is often limited throughout the various growing seasons within Nepal, this is very beneficial.

Timeliness is key when it comes to providing data that will help to care for plants better, thus reducing damage and the potential for resource waste. The smart plant monitoring system was successfully implemented and served as an effective small scale plant monitoring device for a household garden or similar settings.

5. Future Works

There are numerous improvements that can be made to the Smart Plant Monitoring System. AI could be incorporated into future versions to allow users to receive predictive information concerning plants; this could allow users to provide plants with proactive care recommendations. In addition, instead of monitoring a single plant, the system could be utilized for multiple plants through a central dashboard, which would allow large farms and greenhouses to monitor several plants. Furthermore, using solar energy would contribute to making the system more sustainable. In addition, soil pH, nutrient level and other sensory technologies would provide more complete information regarding plant health. The addition of smart irrigation systems (such as drip irrigation) would further conserve water usage, while linking the system to smart home devices or voice activated assistants would enhance the system's usability. These enhancements to the system will create a smarter, more sustainable system that can be used by both home growers and commercial agricultural operations.

- Superior scalability (multi-plant): By building out to allow the system to encompass multiple greenhouses/farms, it becomes more relevant for farms and; therefore, can grow with the customer's needs.
- Renewable Energy: Incorporating solar power into the product line gives farmers/vendors the ability to operate their systems independently from the electrical grid. This could also increase the areas served to Rural Off-Grid Areas or Locations.
- Advanced Sensing: Soil pH/Nutrient/CO_2 sensors would allow users to obtain more accurate, detailed and complete data about plant health.
- Intelligent Irrigation: Automating drip irrigation and/or (Variable Rate) irrigation systems allows growers to continue monitoring and adjusting their irrigation systems according to seasonal variations without constant supervision.

6. Conclusion

The purpose of creating the Smart Plant Monitoring System is to provide an alternative method for maintaining healthy plants by continuously observing the soil and surrounding areas where they grow. This process is more efficient than traditional ways of taking care of plants because many decisions were previously made on the basis of checking the plants manually or estimating their requirements.

Through the use of real-time data, the Smart Plant Monitoring System allows a better understanding of a plant's needs, thereby helping to eliminate problems associated with improper watering (over- or under-watering) and failing to identify environmental changes that affect plant growth and development.

The results from the project demonstrated that through systematic monitoring of plants, resource efficiency is enhanced, and resources are managed more efficiently. In particular, due to the varying climatic conditions and inconsistent access to water that occur within Nepal, an efficient and effective method of utilizing water for plant growth is established, resulting in improved health and vigor of plants.

Through the development of the Smart Plant Monitoring System, it has become evident that structured design, appropriate testing, and reliable integration of the various components used in the development of the monitoring system all contribute to providing consistent and reliable performance in real-world situations.

The project achieved all of its initial goals and objectives successfully, with a practical, adaptable monitoring solution becoming available for all types of home and garden environments (including those growing plants on rooftops), as well as for any small-scale agricultural environment.

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